

John C. Kolesar, Ph.D.

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Full-Time Employment

Amazon Web Services

Applied Scientist

Team: Automated Reasoning Group

Manager: Dominic Mulligan

Portland, Oregon

July 2026–Present

Education

Yale University

Ph.D., Computer Science

Advisor: Ruzica Piskac

Earned M.S. en route to Ph.D. in 2022

New Haven, Connecticut

2020–2026

Cornell University

Bachelor of Arts with Distinction in All Subjects

Majors:

- Mathematics (Magna cum Laude, Computer Science concentration)
- Classics (Latin concentration)

Minors:

- Computer Science
- Philosophy

Cumulative Grade Point Average: 3.97

Computer Science GPA: 4.02

Ithaca, New York

2016–2020

Honors.....

Phi Beta Kappa

Cornell University College of Arts & Sciences

2020

Nathan Hale Associates Fellow

Yale Graduate School of Arts & Sciences

2021

Arts & Sciences Dean's List

Cornell University

All available semesters

Graduate Course Work Performance.....

Grade of H (maximum grade for Yale GSAS) in all graded graduate courses

Thesis

- o John Charles Kolesar. **Coinductive Proof Systems for Private Program Verification.** Doctoral Dissertation, Yale University, 2026.

Research Interests

- Programming Languages
- Formal Verification
- Cryptography
- Zero-Knowledge Proofs
- Symbolic Execution
- Software-Defined Networking
- Automatic Program Repair
- Competitive Programming
- Regular Expressions and KAT

Conference Publications

- Daniel Luick, Ning Luo, John Kolesar, Timos Antonopoulos, Xiao Wang, Eran Tromer, Ruzica Piskac. **Efficient Zero-Knowledge Proofs for First-Order Logic with Quantifiers.** Under Preparation.
- John Kolesar, Timos Antonopoulos, Ruzica Piskac. **Validation of NetKAT Proof Certificates in Zero Knowledge.** Under Submission.
- Jialu Zhang, Jialiang Gu, Wangmeiyu Zhang, José Pablo Cambronero, John Kolesar, Ruzica Piskac, Daming Li, Hanyuan Shi. **Beyond Performance Optimization: Understanding and Repairing Time Limit Exceeded Errors in Online Judge Submissions.** Under Submission.
- John Kolesar, Martin Schäfer, Robin Salkeld, Remy Willems, Willem Visser. **Enforceable Explainability Properties for Pricing Models.** Under Preparation.
- John C. Kolesar, Shan Ali, Timos Antonopoulos, Ruzica Piskac. **Coinductive Proofs of Regular Expression Equivalence in Zero Knowledge.** *OOPSLA*, 2025.
- John Kolesar, Tancrede Lepoint, Martin Schäfer, Willem Visser. **Safe Validation of Pricing Agreements.** *ICSE*, 2025.
- Daniel Luick, John Kolesar, Timos Antonopoulos, William R. Harris, James Parker, Ruzica Piskac, Eran Tromer, Xiao Wang, Ning Luo. **ZKSMT: A VM for Proving SMT Theorems in Zero Knowledge.** *USENIX Security*, 2024.
- John C. Kolesar, Ruzica Piskac, William T. Hallahan. **Checking Equivalence in a Non-strict Language.** *OOPSLA*, 2022.
- Jialu Zhang, De Li, John C. Kolesar, Hanyuan Shi, Ruzica Piskac. **Automated Feedback Generation for Competition-Level Code.** *ASE*, 2022.

Journal Publications

- John C. Kolesar, Ruzica Piskac, William T. Hallahan. **Checking Equivalence in a Non-strict Language.** *Journal of Functional Programming*, 2025.

Talks

ZK Proofs for SMT Theorems and Regular Expression Equivalence

Carnegie Mellon University, CyLab Crypto Seminar

November 2024

Coinductive Proofs of Regular Expression Equivalence in Zero Knowledge

FMCAD 2024 Student Forum

October 2024

ZKSMT: A VM for Proving SMT Theorems in Zero Knowledge

New York University, NJPLS

May 2024

Poster Presentations.....

Coinductive Proofs of Regular Expression Equivalence in Zero Knowledge <i>FMCAD 2024 Student Forum</i>	<i>October 2024</i>
Checking Equivalence in a Non-strict Language <i>Yale University</i>	<i>October 2023</i>

Industry Internships

Amazon Web Services <i>Applied Scientist Intern, New York City</i> Manager: Martin Schäfer Team: AWS Billing	<i>Summer 2024, Summer 2025</i>
Microsoft One Engineering System <i>Research Intern, Remote</i> Supervisor: Josh Becker Mentor: Grant Holliday	<i>Summer 2022</i>
Aretec Inc. <i>Big Data Software Application Developer</i> Contractor for U.S. Securities and Exchange Commission New York City (2018) Washington, D.C. (2019)	<i>Summer 2018, Summer 2019</i>

Mentoring Experience

Zero-Knowledge Regular Expression Equivalence <i>Shan Ali</i>	Yale University <i>Summer 2024</i>
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Teaching Experience

Graduate Teaching Fellow at Yale University.....	
CPSC 415/515: Law and Large Language Models <i>Taught by Ruzica Piskac and Scott Shapiro</i>	<i>Spring 2025</i>
CPSC 458/558: Automated Decision Systems <i>Taught by Stephen Slade</i>	<i>Spring 2024</i>
CPSC 323: Introduction to Systems Programming and Computer Organization <i>Taught by James Glenn and Jay Lim</i>	<i>Fall 2023</i>
CPSC 484/584: Introduction to Human-Computer Interaction <i>Taught by Marynel Vázquez</i>	<i>Spring 2023</i>
CPSC 435/535: Building an Internet Router <i>Taught by Robert Soulé</i>	<i>Fall 2022</i>
CPSC 433/533: Computer Networks <i>Taught by Anurag Khandelwal</i>	<i>Spring 2022</i>
CPSC 323: Introduction to Systems Programming and Computer Organization <i>Taught by Ruzica Piskac and Rob Brunstad</i>	<i>Fall 2021</i>

Undergraduate Teaching Assistant at Cornell University.....

CS 3110: Data Structures and Functional Programming

Taught by Nate Foster

Spring 2020

CS 4820: Introduction to Analysis of Algorithms

Taught by Eva Tardos

Fall 2019

CS 3110: Data Structures and Functional Programming

Taught by Michael Clarkson

Fall 2018

CS 2112: Honors Object-Oriented Design and Data Structures

Taught by Dexter Kozen

Fall 2017

Peer Review Experience

OOPSLA 2024

Artifact Evaluation Committee

Winter, Summer 2024

CAV 2023

Artifact Evaluation Committee

Spring 2023

CAV 2022

Artifact Evaluation Committee

Spring 2022

Other Work, Research, and Volunteering Experience

Cornell University

Computer Science Research

Research Advisor: Nate Foster

Subject: Software-Defined Networking with P4

Ithaca, New York

Fall 2019, Spring 2020

Tenley Achievement Program

Office Manager

Washington, D.C.

Summer 2017

Yale University Computer Science Department

Graduate Student Advisory Committee Member

New Haven, Connecticut

2023–2024 Academic Year

Yale University Computer Science Department

New Ph.D. Student Mentor

New Haven, Connecticut

2023–2024, 2024–2025 Academic Years

National University of Colombia

Web Chair for LPAR 2023

Manizales, Colombia (remote)

June 2023

Squash Haven

Volunteer Tutor (Computer Science, Mathematics)

New Haven, Connecticut

Spring 2023, Fall 2023

Yale University Computer Science Department

Website Manager for Formal Methods Meetup 2023

New Haven, Connecticut

October 2023

Yale University Computer Science Department

Ph.D. Student Buddy for Admitted Student Day

New Haven, Connecticut

Spring 2022

Skills

- o Proficiency in Python, C, C++, Haskell, OCaml, Java, JavaScript, TypeScript, Dafny, Q
- o Experience with Coq, Lean, Isabelle, C#, Standard ML, Kusto, Langium, Rust, Verus
- o Experience with SMT solvers, Excel, LaTeX, Unity, Blender, VirtualBox, Docker